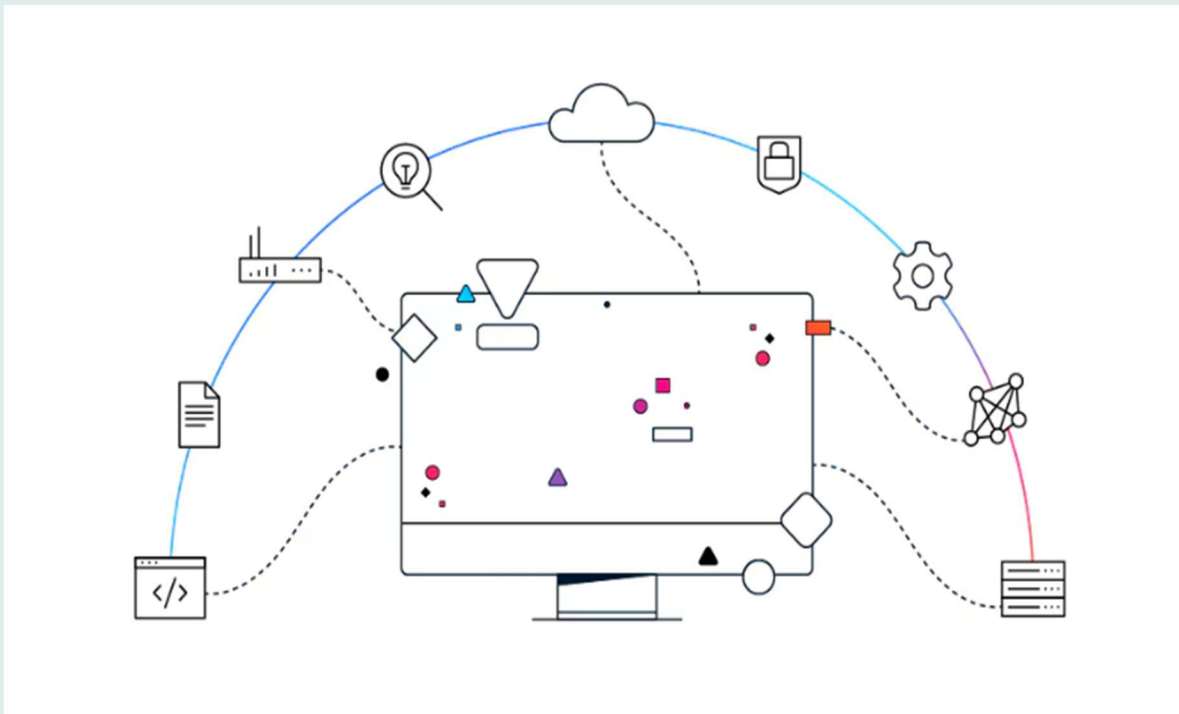


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BE in Computer Science

Enterprise Network Design and Security



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Enterprise Network Design and Security Report

This project presents the design and implementation of a secure enterprise network infrastructure connecting a headquarters location with two branch offices. The network was designed using Cisco Packet Tracer and incorporates VLAN segmentation, Inter-VLAN Routing, DHCP, DNS, OSPF dynamic routing, Access Control Lists (ACLs), and SSH-based secure device management. The architecture focuses on scalability, security, and efficient communication across multiple departments while following industry-standard networking practices.

Modern organizations require reliable and secure communication between multiple offices and departments. Traditional flat networks often suffer from security risks, excessive broadcast traffic, and management difficulties. The challenge was to design a network that ensures secure communication, efficient resource utilization, and simplified administration.

A multi-site enterprise network was developed using Cisco networking technologies. VLAN segmentation was implemented to isolate departments, OSPF was used for dynamic routing between locations, DHCP automated IP allocation, DNS enabled hostname resolution, ACLs enhanced security, and SSH provided secure remote administration.

Enterprise networks require secure, and efficient communication to support modern business operations

Computer networks form the backbone of modern organizations by enabling communication, collaboration, and resource sharing among employees and departments. As organizations expand geographically, maintaining secure, scalable, and efficient communication becomes increasingly important. Enterprise networks must support reliable data transmission, secure access to resources, and effective network management while ensuring security, performance, and business continuity. This project focuses on designing and implementing a secure enterprise network that enables seamless connectivity between headquarters and branch offices using industry-standard networking technologies and best practices.

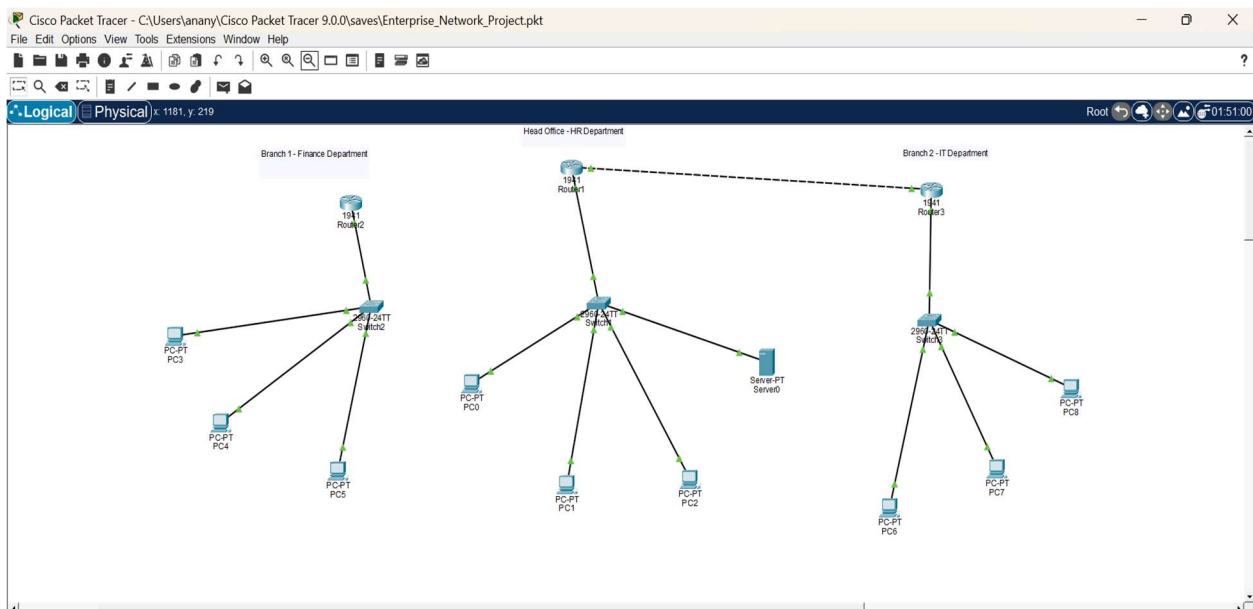
Project Scope

This project focuses on the design and implementation of a secure enterprise network infrastructure connecting a headquarters office with two branch offices. The network was developed using Cisco Packet Tracer and incorporates industry-standard networking technologies including VLANs, Inter-VLAN Routing, DHCP, DNS, OSPF, Access Control Lists (ACLs), and SSH.

The architecture is designed to provide secure communication, efficient traffic management, network scalability, and centralized administration. The project simulates a real-world enterprise environment where different departments are logically separated while maintaining controlled communication between network segments.

Project Objectives

- *Design a scalable enterprise network architecture.*
- *Implement VLAN-based segmentation.*
- *Configure Inter-VLAN Routing.*
- *Implement DHCP services.*
- *Configure DNS services.*
- *Deploy OSPF dynamic routing.*
- *Secure communication using ACLs.*
- *Configure SSH for secure administration.*
- *Validate network performance through testing.*



Network Components

DEVICE	QUANTITY
Router	3
Switches	3
PCs	12
DNS Server	1

IP Addressing Plan

NETWORK SEGMENT	ADDRESS
HR VLAN	192.168.10.0/24
Finance VLAN	192.168.20.0/24
IT VLAN	192.168.30.0/24
Branch 1	192.168.40.0/24
Branch 2	192.168.50.0/24

VLAN Implementation

```
Switch>
Switch>enable
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
10	HR	active	Fa0/1, Fa0/4
20	Finance	active	Fa0/2
30	IT	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

[Copy](#)[Paste](#)[Top](#)

VLANs were configured to logically separate departments and reduce unnecessary broadcast traffic. The implementation improves security and allows efficient traffic management by grouping users based on business functions.

VLAN	DEPARTMENT
10	HR
20	Finance
30	IT

Inter - VLAN Routing

```

PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.50

Pinging 192.168.10.50 with 32 bytes of data:

Reply from 192.168.10.50: bytes=32 time<1ms TTL=128
Reply from 192.168.10.50: bytes=32 time=1ms TTL=128
Reply from 192.168.10.50: bytes=32 time<1ms TTL=128
Reply from 192.168.10.50: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

Router-on-a-Stick configuration was implemented to enable communication between VLANs while maintaining network segmentation. This allows departments to share resources without compromising security.

DHCP and DNS Services

```

HQ-Router>enable
HQ-Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
                Hardware address
192.168.10.2    0005.5E9E.D4C2    --                  Automatic
HQ-Router#

```

DHCP

DHCP automatically assigns IP addresses to devices joining the network, reducing administrative overhead and configuration errors.

DNS

DNS services translate hostnames into IP addresses, allowing users to access network resources using easy-to-remember names.

DNS Record

```
C:\>ping www.company.local

Pinging 192.168.10.50 with 32 bytes of data:

Reply from 192.168.10.50: bytes=32 time<1ms TTL=128
Reply from 192.168.10.50: bytes=32 time<1ms TTL=128
Reply from 192.168.10.50: bytes=32 time=12ms TTL=128
Reply from 192.168.10.50: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.50:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

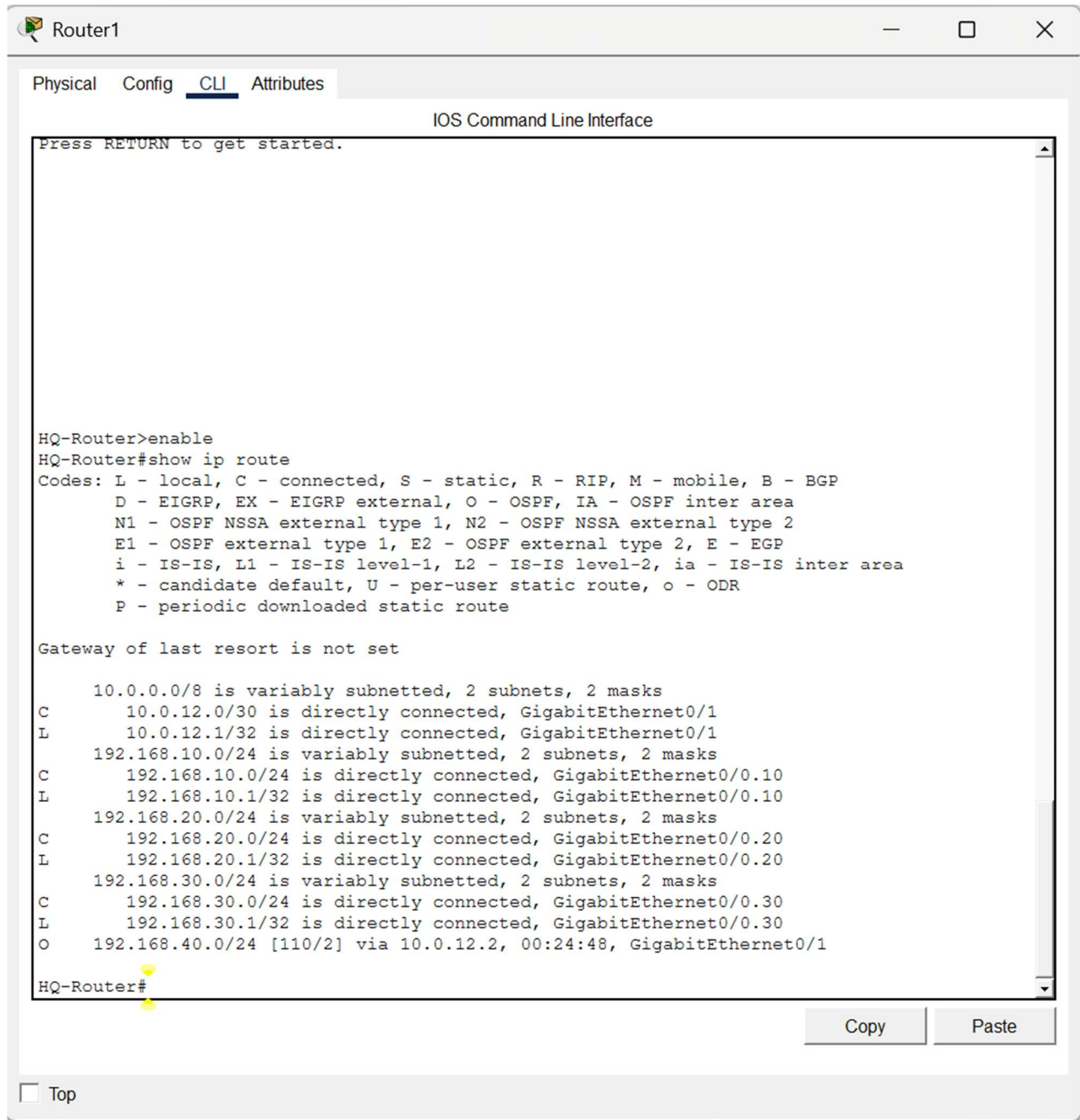
OSPF Dynamic Routing

OSPF Neighbor:

```
HQ-Router>
HQ-Router>enable
HQ-Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.40.1	1	FULL/DR	00:00:36	10.0.12.2	GigabitEthernet0/1

☐ Top



OSPF was selected as the routing protocol due to its scalability and rapid convergence. The protocol automatically exchanges routing information among routers, ensuring optimal path selection across the enterprise network.

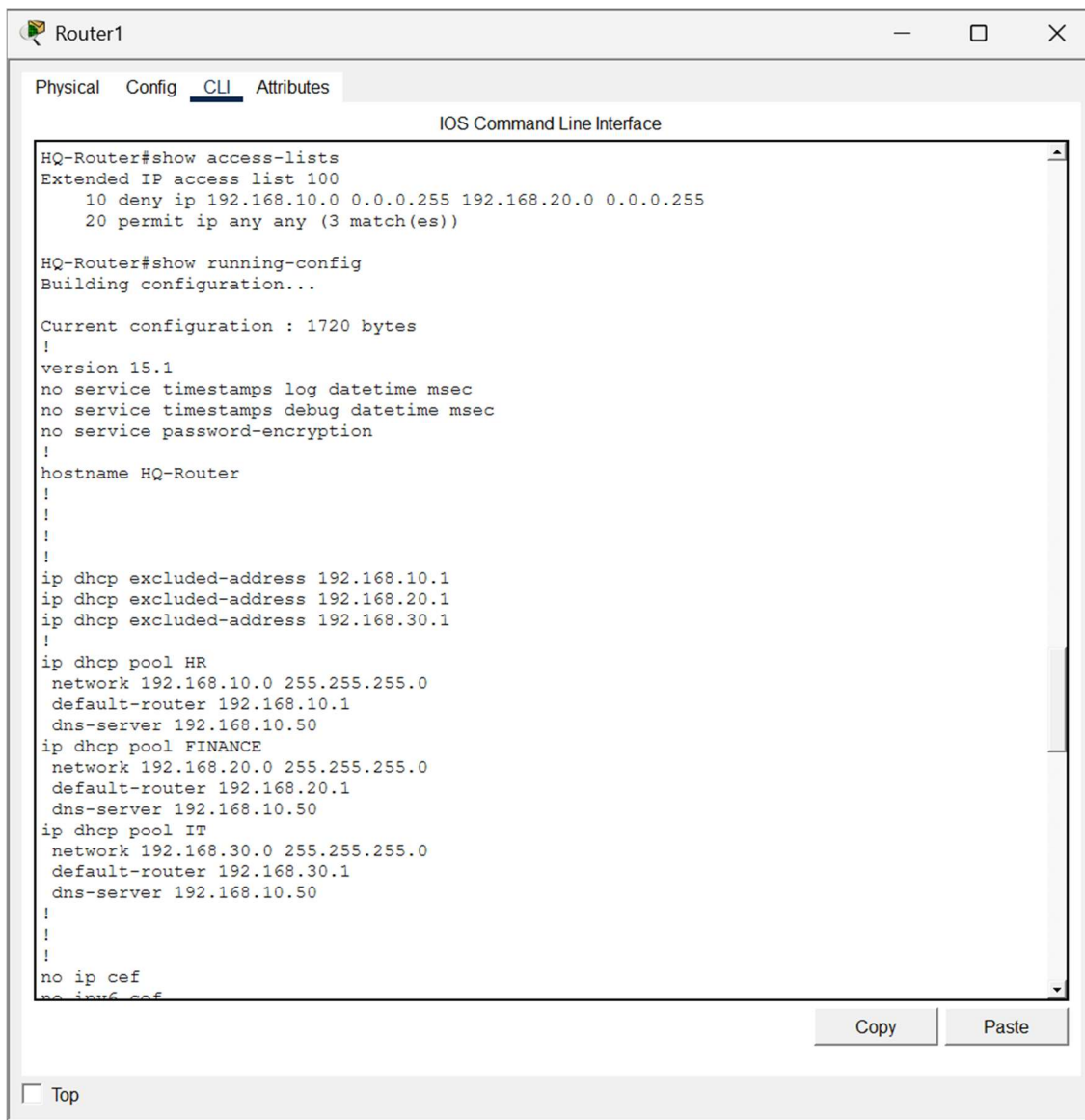
Network Security Implementation

ACLs

Access Control Lists were used to restrict unauthorized communication between departments.

SSH

SSH was configured to enable encrypted remote management of network devices.



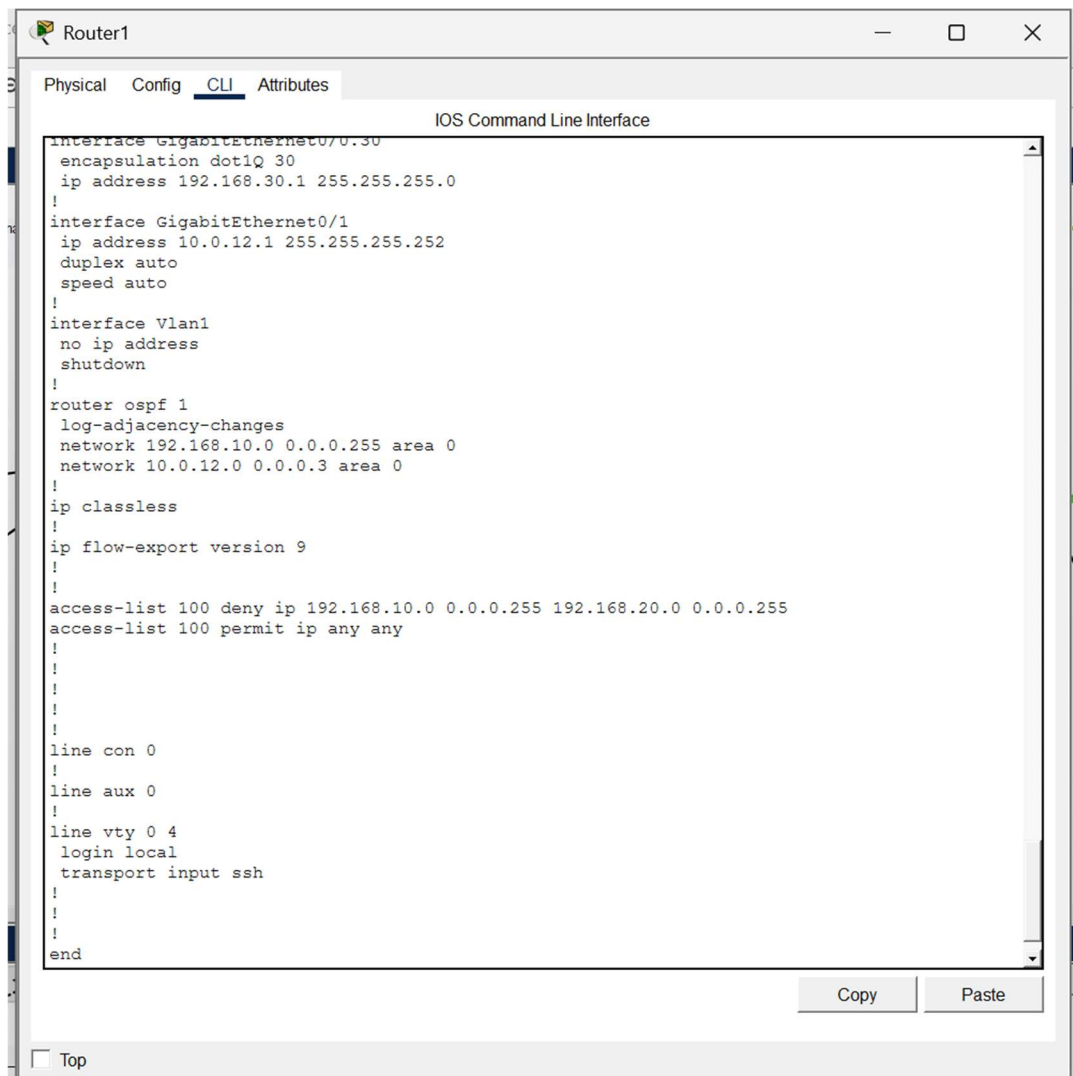
The screenshot shows a Cisco Router CLI window titled "Router1". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area displays the "IOS Command Line Interface" with the following text:

```
HQ-Router#show access-lists
Extended IP access list 100
  10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
  20 permit ip any any (3 match(es))

HQ-Router#show running-config
Building configuration...

Current configuration : 1720 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname HQ-Router
!
!
!
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
!
ip dhcp pool HR
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.10.50
ip dhcp pool FINANCE
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 192.168.10.50
ip dhcp pool IT
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 192.168.10.50
!
!
!
no ip cef
no ip vrf def
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.



Testing and Results

ITEM	NEEDED
VLAN Communication	Success
DHCP Assignment	Success
DNS Resolution	Success
OSPF Routing	Success
SSH Connectivity	Success
ACL Enforcement	Success

Skills Demonstrated

NETWORKING
IP Addressing
Subnetting
Routing & Switching
OSI Model
TCP/IP

CISCO TECHNOLOGIES
VLAN (Virtual Local Area Network)
Trunking
Router-on-a-Stick
OSPF (Open Shortest Path First)
DHCP (Dynamic Host Configuration Protocol)
DNS (Domain Name System)

SECURITY
Access Control Lists (ACLs)
SSH (Secure Shell)
Network Segmentation

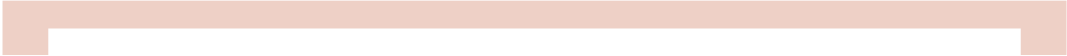
PROFESSIONAL SKILLS
Network Documentation
Problem Solving
Troubleshooting
Systematic Testing

Challenges and Solutions

CHALLENGE	SOLUTION
VLAN Misconfiguration	Verified switch port assignments
OSPF Neighbors Not Forming	Corrected OSPF network statements
DHCP Issues	Reconfigured DHCP pools
ACL Blocking Traffic	Modified ACL sequence

Conclusion

The project successfully demonstrated the design and implementation of an enterprise-grade multi-site network infrastructure. Through the integration of routing, switching, dynamic routing protocols, network services, and security mechanisms, the solution achieved reliable communication, enhanced security, and scalability. The project provided hands-on experience with Cisco technologies and strengthened practical networking skills applicable to enterprise environments.



Designing secure and scalable enterprise networks requires the integration of routing, switching, security, and network services to support modern business operations.

